DPEN022: Enabling Mathematics B
Class Questions Week 4

Week 4A

Key Idea – Derivatives of Logarithmic Functions

1. Differentiate the following:

(a) $y = \log_e 4x$ (b) $y = 3\log_e(6x+5) + 2x$ (c) $f(x) = \log_e(2x^4+1)$

(d) $y = \ln(4x+3) - 2x$ (e) $y = 2\log_e x + 3x^2 - 2$ (f) $f(x) = 2\ln(x^2 - 4x + 4)$

(g) $f(x) = \log_e(1+\sqrt{x})$ (h) $y = \ln x^4 + 3x$ (i) $y = \log_e\left(\dfrac{x}{2}\right)$

2. Find the gradient function of the following *[using log laws]*:

(a) $y = \log_e(x+2)(x-5)$ (b) $f(x) = \log_e\sqrt{(x^2+4)}$ (c) $f(x) = \log_e\dfrac{(x+2)}{(x-1)}$

(d) $g(x) = \log_e\dfrac{2}{(4x-3)}$ (e) $f(x) = \log_e 2x(x^2-1)$ (f) $y = \log_e(3x^2+1)^2$

3. Find the derivative of the following *[using product, quotient and chain rule]*:

(a) $y = 2x^3\ln x$ (b) $y = (x+3)\ln(x+3)$ (c) $y = \dfrac{\log_e(x^2+1)}{x}$

(d) $f(x) = \left[\log_e(x^3+2)\right]^4$ (e) $g(x) = 1 + 2x\log_e x$ (f) $y = 2x - [\log_e(x+2)]^3$

4. (a) Given $g(x) = x^2\log_e x$ find $g'(1)$.

(b) If $y = x\log_e(x) - x$ find y''.

(c) Find the point on the curve $y = \dfrac{\ln x}{x}$ where the gradient is equal to zero.

(d) Find the equation of the tangent to the curve $f(x) = \log_e(x-1)$ when $x = 2$.

Write the equation in the general form.

Key Idea – Derivatives of Trigonometric Functions

1. Differentiate the following:

(a) $y = \sin 3x$ (b) $f(x) = \sin x + 4\cos x$ (c) $y = \tan x^2$

(d) $y = 2\sin(2x)$ (e) $y = \tan(4x-1)$ (f) $y = 4\cos\left(x + \frac{\pi}{3}\right)$

(g) $f(x) = 3\sin 2x - \cos 2x$ (h) $y = \tan\left(\frac{\pi}{4} + 4x\right) - 2x^2$ (i) $y = \sin\frac{x}{2} - x^3$

2. Differentiate the following:

(a) $y = x^2\sin(2x)$ (b) $y = \sqrt{(1+\tan(3x))}$

(c) $y = \sin^2(2x+1)$ (d) $f(x) = \dfrac{x^2}{\cos x}$

(e) $f(x) = x\cos\dfrac{x}{2}$ (f) $f(x) = \sin x(1+\cos x)$

(g) $y = \dfrac{x^3}{\cos x}$ (h) $y = \dfrac{x}{\tan(2x)}$

3. (a) Given $f(x) = 2\cos(3x)$, find $f'\left(\frac{\pi}{12}\right)$.

(b) If $y = 2\sin x - \cos(4x-1)$ find $\dfrac{d^2y}{dx^2}$.

(c) Find the equation of the tangent to the curve $y = \tan x$ when $x = \dfrac{\pi}{3}$.

Key Idea – Mixed Derivative of any Functions using Rules 1-5

1. Differentiate with respect to x.

(a) $y = e^x\sin x$ (b) $y = e^{2x}\cos\dfrac{x}{2}$ (c) $y = e^{-x}\sin 3x$

(d) $f(x) = e^x\cos 4x$ (e) $y = (\cos x + \sin x)e^{-x}$ (f) $f(x) = e^{\sin 2x}$

(g) $f(x) = e^x\ln x$ (h) $y = \dfrac{\log_e x}{e^x}$ (i) $y = e^x\log_e(e^x+1)$

2. Differentiate the following:

(a) $f(x) = (2x - 3e^x + 4\tan x)^{11}$ (b) $f(x) = e^{x^2+\tan(5x)}$

(c) $y = \tan(\sin x + \cos x)$ (d) $y = \cos(5e^{5x} + x)$

(e) $y = \cos x\tan x$ (f) $y = \ln(5x + \sin(x^2))$

(g) $y(x) = \ln((\cos x)^{-4})$ (h) $f(x) = e^{x\ln x}$

Week 4D

Key Idea –The Primitive Function

1. The gradient of a curve is given by $\frac{dy}{dx} = 6 - 4x$. What is the equation of the curve if it passes through the point (1, 7)? $y = 3 + 6x - 2x^2$.
2. The gradient of a curve is given by $\frac{dy}{dx} = 6x^2 + 8x$. If the curve passes through the point (1, -3), find the equation of the curve. $y = 2x^3 + 4x^2 - 9$.

Key Idea – Integration of Power of *x* where *n* is an Integer

1. Find the primitive function of the following:

(a) $\frac{dy}{dx} = x^3$	(b) $\frac{dy}{dx} = 6$	(c) $\frac{dy}{dx} = 2x^4$
(d) $\frac{dy}{dx} = -6x^2$	(e) $\frac{dy}{dx} = \frac{4}{5}x^3$	(f) $\frac{dy}{dx} = \frac{x^2}{5}$
(g) $\frac{dy}{dx} = x^{-3}$	(h) $\frac{dy}{dx} = -2x^{-4}$	(i) $\frac{dy}{dx} = \frac{2}{3}x^{-6}$
(j) $\frac{dy}{dx} = \frac{1}{x^4}$	(k) $\frac{dy}{dx} = \frac{-3}{x^5}$	(l) $\frac{dy}{dx} = \frac{2}{3x^3}$

Key Idea – Integration of Power of *x* where *n* is a Fraction

1. Find the following, writing your answer in positive index form and surd form:

(a) $\int x^{\frac{1}{4}}\, dx$	(b) $\int 2x^{\frac{2}{5}}\, dx$	(c) $\int -3x^{\frac{2}{3}}\, dx$
(d) $\int x^{-\frac{1}{4}}\, dx$	(e) $\int 8x^{-\frac{3}{4}}\, dx$	(f) $\int \frac{2}{3x^{\frac{2}{3}}}\, dx$
(g) $\int \sqrt[3]{x^2}\, dx$	(h) $\int \sqrt{3x}\, dx$	(i) $\int -7\sqrt[4]{x}\, dx$
(j) $\int \frac{1}{\sqrt{x}}\, dx$	(k) $\int \frac{-2}{3\sqrt{x}}\, dx$	(l) $\int \frac{1}{5\sqrt{x^3}}\, dx$
(m) $\int x\sqrt{x}\, dx$	(n) $\int 4x^2\sqrt{x}\, dx$	(o) $\int \frac{x^2}{\sqrt{x}}\, dx$

Key Idea - Integration of Power of *x* where *n* is an Integer or Fraction

1. Find the following indefinite integrals, writing your answer in positive index form or surd form:

(a) $5x + x^2 - 3x^3$	(b) $x^2 - 1$	(c) $x^5 - 4x^3 - 2x^2 + 1$
(d) $\frac{x^2}{5} - \frac{x^3}{4}$	(e) $\sqrt{x} + \sqrt[2]{x}$	(f) $-2x^4 - \frac{1}{2x^2}$
(g) $\frac{5}{x^2} + 2x^{\frac{1}{2}} + 3$	(h) $3 + \frac{1}{x^2} - \frac{2}{x^3}$	(i) $\frac{1}{\sqrt{x}} + \frac{1}{x\sqrt{x}}$

Key Idea - Integration by Simplifying the Integrand

1. Integrate the following, writing your answer in positive index form:

(a) $\int 2x^2\left(3x+\frac{1}{x^2}\right)\,dx$

(b) $\int \left(x^2-5\right)\left(x^3-7\right)\,dx$

(c) $\int \frac{2x^3-2x^5+x^9}{x^3}\,dx$

(d) $\int 2x^3\left(3x^2-1\right)\,dx$

(e) $\int \frac{-2+x^4}{x^3}\,dx$

(f) $\int \sqrt{x}\left(1+\frac{1}{\sqrt{x}}\right)\,dx$

(g) $\int (2x+1)^2\,dx$

(h) $\int \left(x^2-1\right)\left(x^2+1\right)\,dx$

Mixed

a) Integrate the following with respect to x:

i) $4x^3 - 3x^{-4} + 8x - 1$

ii) $\dfrac{3x^3 - 2}{x^2} + \sqrt{x}$

iii) $(5x^2 + 6)^2$

a) i) $x^4 + x^{-3} + 4x^2 - x + c$ ii) $\dfrac{3}{2}x^2 + 2x^{-1} + \dfrac{2}{3}x^{\frac{3}{2}} + c$

iii) $5x^5 + 20x^3 + 36x + c$

i) $5x^{-4} - \dfrac{x}{2} + \sqrt[4]{x}$ (3)

ii) $\dfrac{2x^4 - 4x^3 + 3}{x^3}$ (2)

(2)

iv) $(3x-1)(3x+1)$ (2)

6. (a) Find the following (indefinite) integrals:

(i) $\int x^3 + 4x + 1 \, dx$ (ii) $\int (5 - 2x)^4 \, dx$

(iii) $\int \dfrac{x+4}{x^3} \, dx$ (iv) $\int (3x-1)(x^2-6) \, dx$

6. (a) (i) $\dfrac{x^4}{4} + 2x^2 + x + c$ (ii) $\dfrac{(5 - 2x)^5}{-10} + c$

(iii) $-\dfrac{1}{x} - \dfrac{2}{x^2} + c$ (iv) $\dfrac{3x^4}{4} - 9x^2 - \dfrac{x^3}{3} + 6x + c$

1) $6x^2 - x$

2) $(x+1)(x-3)$

3) $\dfrac{1}{x^8} + \sqrt{x}$

41) $2x^3 - \dfrac{x^2}{2} + c$

42) $\dfrac{x^3}{3} - x^2 - 3x + c$

43) $-\dfrac{1}{7x^7} + \dfrac{2}{3}x^{\frac{3}{2}} + c$